Slide Set #11

Model Specification and Identification

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Notes

Part I
Specification

Notes

Specification

Specification
is the *art* of finding: (i) the set of regressors that explain/predict the Y *appropriately*; (ii) and *appropriate* settings for the error term

Variable selection
choice of the *relevant* set of regressors. It is the central issue of model specification although choosing a model means more

Predictive paradigm
easier situation, because it is possible to assess the main goal (prediction error). However, even in this case we can get easily in the *overfitting* trap.

Explanatory/Causal paradigm
much harder, nobody really knows when it is done right! It relies on a combination of theoretical insights and empirical experience. In general we try to exclude causes of bad specification

Notes

Variable Selection

Consider two competing models

Model 1: $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

Model 2: $\boldsymbol{y} = \boldsymbol{X}_0\boldsymbol{\theta}_0 + \boldsymbol{X}\boldsymbol{\theta} + \boldsymbol{\nu}$

Sample problem from two different perspectives

1. Predictive paradigm: which model better predicts Y
2. Explanatory/Causal paradigm: which model explains accurately the relationships between Y and Xs? Which model is "*closer to the true*" mechanism producing observed data?

Notes

Overifitting trap

Consider the OLS fits of the previous models

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{b} + \boldsymbol{e} \tag{11.1}$$

and

$$\boldsymbol{y} = \boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{X}\boldsymbol{t} + \boldsymbol{v} \tag{11.2}$$

From which

$$\boldsymbol{X}\boldsymbol{b} + \boldsymbol{e} \quad = \quad \boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{X}\boldsymbol{t} + \boldsymbol{v} \tag{11.3}$$

Since these are all OLS fits, then $\boldsymbol{e}'\boldsymbol{X} = \boldsymbol{v}'\boldsymbol{X}_0 = \boldsymbol{v}'\boldsymbol{X} = 0$.

Notes

Multiply both sides of (11.3) by $\boldsymbol{e}'$ and $\boldsymbol{v}'$, and obtain

$$\begin{aligned} \boldsymbol{e}'\boldsymbol{e} =& \boldsymbol{e}'\boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{e}'\boldsymbol{v} \\ \boldsymbol{v}'\boldsymbol{e} =& \boldsymbol{v}'\boldsymbol{v} \end{aligned}$$

which implies

$$\boldsymbol{e}'\boldsymbol{X}_0\boldsymbol{t}_0 = \boldsymbol{e}'\boldsymbol{e} - \boldsymbol{v}'\boldsymbol{v} \tag{11.4}$$

Now we show that going from the first OLS fit to the second, the sum of squared residuals cannot increase, that is: $\boldsymbol{v}'\boldsymbol{v} \leq \boldsymbol{e}'\boldsymbol{e}$

Notes

$$
\begin{aligned}
\boldsymbol{v}'\boldsymbol{v} &= (\boldsymbol{y} - \boldsymbol{X}_0\boldsymbol{t}_0 - \boldsymbol{X}\boldsymbol{t})'(\boldsymbol{y} - \boldsymbol{X}_0\boldsymbol{t}_0 - \boldsymbol{X}\boldsymbol{t}) \\
&= (\boldsymbol{X}\boldsymbol{b} + \boldsymbol{e} - \boldsymbol{X}_0\boldsymbol{t}_0 - \boldsymbol{X}\boldsymbol{t})'(\boldsymbol{X}\boldsymbol{b} + \boldsymbol{e} - \boldsymbol{X}_0\boldsymbol{t}_0 - \boldsymbol{X}\boldsymbol{t}) \\
&= (\boldsymbol{X}(\boldsymbol{b} - \boldsymbol{t}) - \boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{e})'(\boldsymbol{X}(\boldsymbol{b} - \boldsymbol{t}) - \boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{e}) \\
&\text{set } \boldsymbol{a} = \boldsymbol{X}(\boldsymbol{b} - \boldsymbol{t}) - \boldsymbol{X}_0\boldsymbol{t}_0 \\
&= \boldsymbol{a}'\boldsymbol{a} + 2\boldsymbol{e}'\boldsymbol{a} + \boldsymbol{e}'\boldsymbol{e} \\
&= \boldsymbol{a}'\boldsymbol{a} - 2\boldsymbol{e}'\boldsymbol{X}_0\boldsymbol{t}_0 + \boldsymbol{e}'\boldsymbol{e} \\
&= \boldsymbol{a}'\boldsymbol{a} - 2(\boldsymbol{e}'\boldsymbol{e} - \boldsymbol{v}'\boldsymbol{v}) + \boldsymbol{e}'\boldsymbol{e} \\
&= \boldsymbol{a}'\boldsymbol{a} - \boldsymbol{e}'\boldsymbol{e} + 2\boldsymbol{v}'\boldsymbol{v}
\end{aligned}
$$

Observe that $\boldsymbol{v}'\boldsymbol{v} = \boldsymbol{e}'\boldsymbol{e} - \boldsymbol{a}'\boldsymbol{a}$ implies that $\boldsymbol{v}'\boldsymbol{v} \leq \boldsymbol{e}'\boldsymbol{e}$

Main conclusion

adding more regressors will reduce the RSS and increases the R^2 (this is why is better to look at the adjusted-R^2)

Notes

Predictive paradigm

Goal

predict Y as good as possible. A low prediction error is the targeted

Overfitting

whenever the model becomes richer/more flexible (adding regressors), it will fit the observed data not worse. Model adapts too much to the observed sample and is not able to predict *unseen* data

Aproaches to solve the problem

- Penalized (in-sample) measures of fit: an *in-sample* statistics of fit that *penalizes* for the addition of regressors
- Out-of-sample prediction error: estimate the expected prediction error on samples not used to fit the model

Notes

Penalized (in-sample) measures of fit

Adjusted R^2 (to be maximized)

$$\overline{R}^2 = 1 - \frac{n-1}{n-K}(1-R^2)$$

It is not based on distributional assumptions for the error term

Akaike information criterion (to be minimized)

$$AIC = \log\left(\frac{\text{RSS}}{n}\right) + \frac{2K}{n}$$

Needs assumptions on the error distribution

Bayes information criterion (to be minimized)

$$BIC = \log\left(\frac{\text{RSS}}{n}\right) + \frac{K\log(n)}{n}$$

Needs assumptions on the error distribution

Notes

Out-of-sample prediction error via cross-validation

Step 1: Fix $H < n$ and divide the n units into two subsets:

- Train set: randomly(*) select $n - H$ units
- Test set: select the remaining H units. Let $y_1^o, y_2^o, \ldots, y_H^o$ the Y-values in this set.

Step 2: In-sample estimation
apply the OLS to estimate $\boldsymbol{b}$ using observations in the Train set

Step 3: Out-of-sample prediction
predict the Y-values on the Test set using $\boldsymbol{b}$. Let $\hat{y}_1^o, \hat{y}_2^o, \ldots, \hat{y}_H^o$ the predicted values

Remark(*): the appropriate random split depends on the sampling design

Notes

Step 4: measure the expected prediction error
The most popular prediction accuracy measure is the *root mean square prediction error*

$$\text{RMSPE} = \sqrt{\frac{1}{H}\sum_{h=1}^{H}(y_h^o - \hat{y}_h^o)^2}$$

Another popular measure is the *mean absolute prediction error*

$$\text{MAE} = \frac{1}{H}\sum_{h=1}^{H}|y_h^o - \hat{y}_h^o|$$

Since the splitting introduces additional random variations, steps 1–4 are repeated a number of times and the resulting RMSPE or MAE values are averaged across splittings.

Notes

Explanatory/causal paradigm

Goal

assess the structural relationships between X and Y, produce "*good estimates*" and *valid inference* about partial effects.

Example:

$$\texttt{log(wage)}_i = \beta_1 + \beta_2 \texttt{SchoolYears}_i + \ldots + \varepsilon_i$$

- we want an accurate estimate of β_2, because we need the expected variation of `log(wage)` produced by a unit change of `SchoolYears`
- we want that $\mathrm{Se}(b_2)$ is accurate
- we want that rejection regions in hypothesis tests are valid (and optimal)

Relevant aspects

we are interested in how the model misspecification impact the quality of estimates and the resulting inference

Notes

Effects of misspecification on estimation/inference

Reconsider the two alternative models

Model $(m = 1)$: $\quad \boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$

Model $(m = 2)$: $\quad \boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{u}$

2 Cases

- *inclusion of irrelevant variables:* truth = Model 1, but we estimate Model 2?
- *omission of relevant variables:* truth = Model 2, but we estimate Model 1?

Notations: there are 2 parameters vectors $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$. Let $\boldsymbol{b}_h^{(m)}$ be the OLS estimate of $\boldsymbol{\beta}_h$ based on Model (m). Let $\boldsymbol{b}^{(m)}$ be the overall OLS estimator under Model (m)

Notes

Inclusion of irrelevant variables

Assume Model 1 generates data (true model), but we estimate Model 2. We don't realized that $\boldsymbol{\beta}_2 = \mathbf{0}$

Consequences

- the OLS is still unbiased and consistent for $\boldsymbol{\beta}_1$

$$\begin{array}{l} \mathrm{E}[\boldsymbol{b}_1^{(2)} \mid \boldsymbol{X}_1, \boldsymbol{X}_2] = \boldsymbol{\beta}_1 \\ \mathrm{E}[\boldsymbol{b}_2^{(2)} \mid \boldsymbol{X}_1, \boldsymbol{X}_2] = \mathbf{0} \end{array} \qquad \text{and} \qquad \begin{pmatrix} \boldsymbol{b}_1^{(2)} \\ \boldsymbol{b}_2^{(2)} \end{pmatrix} \xrightarrow{\mathsf{p}} \begin{pmatrix} \boldsymbol{\beta}_1 \\ \mathbf{0} \end{pmatrix}$$

- s^2 is unbiased for σ^2
- $\widehat{\mathrm{AVar}}\left(\boldsymbol{b}^{(2)}\right)$ is consistent
- Loss of efficiency: $\mathrm{Var}[\boldsymbol{b}_1^{(2)} \mid \boldsymbol{X}_1, \boldsymbol{X}_2] \succeq \mathrm{Var}[\boldsymbol{b}_1^{(1)} \mid \boldsymbol{X}_1] \implies$. The loss increases with the correlation between $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$
- Test power reduces: because probability of false negatives increases (not rejecting H_0 when H_1 is true). E.g. the default test tempts to *exclude relevant variables*

Notes

Omission of relevant variables

Assume Model 2 = truth, but we estimate estimate Model 1. Assume $\beta_2 \neq \mathbf{0}$, that is $\boldsymbol{X}_2$ correlates with $\boldsymbol{y}$

Observe that

- from true Model 2: $\boldsymbol{y} - \boldsymbol{X}_1\beta_1 = \boldsymbol{X}_2\beta_2 + \boldsymbol{u}$
- from wrong Model 1: $\varepsilon = \boldsymbol{y} - \boldsymbol{X}_1\beta_1$

Therefore $\varepsilon = \boldsymbol{X}_2\beta_2 + \boldsymbol{u}$

$\boldsymbol{X}_2$ is called "*unobserved heterogeneity*"
variations across individual units not accounted by the specified model, and that will go into the error term of the misspecified model

Endogeneity
if $\boldsymbol{X}_2$ correlates with $\boldsymbol{X}_1$, error of the fitted model correlates with the regressors violating orthogonality. Failure of orthogonality leads to non identifiable models (see later).

⚠ **EFFECTS OF ENDOGENEITY ARE DEVASTATING** ⚠

Notes

For exampl, it is easy to see that OLS is not even unbiased

In fact, $\boldsymbol{b}_1^{(1)} = (\boldsymbol{X}_1{}'\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1{}'\boldsymbol{y}$ can be rewritten as

$$\boldsymbol{b}_1^{(1)} = (\boldsymbol{X}_1{}'\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1{}'(\boldsymbol{X}_1\boldsymbol{\beta}_1 + \boldsymbol{X}_2\boldsymbol{\beta}_2 + \boldsymbol{u})$$

that is

$$\boldsymbol{b}_1^{(1)} = \boldsymbol{\beta}_1 + (\boldsymbol{X}_1{}'\boldsymbol{X}_1)^{-1}(\boldsymbol{X}_1{}'\boldsymbol{X}_2)\boldsymbol{\beta}_2 + (\boldsymbol{X}_1{}'\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1{}'\boldsymbol{u}$$

Assuming that strict esogeneity holds for the true model, i.e. $\mathrm{E}[\boldsymbol{X}_1{}'\boldsymbol{u}|\boldsymbol{X}_1, \boldsymbol{X}_2] = \mathbf{0}$, then

$$\mathrm{E}[\boldsymbol{b}_1^{(1)} \mid \boldsymbol{X}_1, \boldsymbol{X}_2] = \boldsymbol{\beta}_1 + (\boldsymbol{X}_1{}'\boldsymbol{X}_1)^{-1}(\boldsymbol{X}_1{}'\boldsymbol{X}_2)\boldsymbol{\beta}_2$$

Therefore $\boldsymbol{b}_1^{(1)}$ is **biased** unless $\boldsymbol{X}_1{}'\boldsymbol{X}_2 = \mathbf{0}$ (note: by assumption $\beta_2 \neq \mathbf{0}$)

Notes

Consequences of omission of relevant variables

If the omitted variables correlates with the included regressors we observe several nasty effects:

- biased and inconsistent OLS estimate of β_1. This is caused by the lack of identifiability (see later)
- s^2 is biased, it under-estimates the error variance.
- $\widehat{\mathrm{AVar}}\left(\boldsymbol{b}_1^{(1)}\right)$ is not consistent, and standard errors are under-estimated so precision of estimated coefficients *apparently improves*
- No guarantees about hypothesis tests and confidence intervals

Notes

Part II
Identification

Notes

Identification

The distribution function $F(y;\theta)$ is your probability model for an observable quantity of interest y. F is known if its parameter θ is known.

We have a sample $\{y_1, y_2, \ldots, y_n\}$, and we want to recover the θ from the sampling information.

Suppose that there are $\theta_1 \neq \theta_2$ such that $F(y;\theta_1) = F(y;\theta_2)$.
⚠ This means that the sample is consistent with two different parameters!

Estimation makes sense if different parameters will produce different distributions. This requirement is called identifiability of the model

Motto:

NINE = No Identification No Estimation

Notes

Definition (global identifiability)

The model $F(\,\cdot\,;\theta)$ is identifiable if for any pair θ', θ'' taken in the parameter space: $\theta' \neq \theta'' \implies F(\,\cdot\,;\theta') \neq F(\,\cdot\,;\theta'')$

Remark: consistency doesn't make sense for non identifiable models

- $\hat{\theta}$ is consistent for θ': it means that for large n is "*rather likely*" that $\hat{\theta}$ belongs to a small neighbourhood of θ'.
- suppose that θ' and θ'' produce the same data distribution. Then as $n \to +\infty$ it is "*rather likely*" that $\hat{\theta}$ keep switching between two small neighbourhoods of θ' and θ''.... so it can't converge!

There are weaker versions of the identifiability concept. Not important here.

Proving identifiability is "art".

Notes

Identification for the scalar linear regression model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Assume we know the joint distribution of the observables (Y, X). In that case we are able to compute moments, e.g. $\mathrm{E}[Y]$, $\mathrm{E}[Y]$, $\mathrm{Var}[X]$, and $\mathrm{Cov}[Y, X]$

Are (β_0, β_1) identifiable? Note that

$$\begin{aligned}\mathrm{Cov}[Y, X] &= \mathrm{Cov}[\beta_0 + \beta_1 X + \varepsilon, X] \\ &= \mathrm{Cov}[\beta_0, X] + \beta_1 \mathrm{Cov}[X, X] + \mathrm{Cov}[X, \varepsilon] \\ &= \beta_1 \mathrm{Var}[X] + \mathrm{Cov}[X, \varepsilon]\end{aligned}$$

A sufficient condition for identification of β_1
assume $\mathrm{Cov}[X, \varepsilon] = 0$ and $\mathrm{Var}[X] > 0$, then

$$\beta_1 = \frac{\mathrm{Cov}[Y, X]}{\mathrm{Var}[X]}$$

Now β_1 is identified. Why?

Notes

Observe

$$\mathrm{E}[Y] = \beta_0 + \beta_1 \mathrm{E}[X] + \mathrm{E}[\varepsilon]$$

A sufficient condition to identify β_0
assume $\mathrm{E}[\varepsilon] = 0$. Then

$$\beta_0 \;=\; \mathrm{E}[Y] - \beta_1 \mathrm{E}[X] \;=\; \mathrm{E}[Y] - \frac{\mathrm{Cov}[Y, X]}{\mathrm{Var}[X]} \mathrm{E}[X]$$

Now β_0 is identified. Why?

Remarks

- the two crucial assumptions here require exogeneity. Orthogonality between errors and regressors is crucial to achieve exogeneity.
- that's why these type of assumptions is common to all regression models. They lead to identifiability
- another the trivial consequence of establishing idenitifiability here is that any set of estimators consistent for $\mathrm{E}[X]$, $\mathrm{E}[Y]$, $\mathrm{Var}[X]$, and $\mathrm{Cov}[X, Y]$, will asymptotically *uniquely* recover β_0 and β_1

Notes

Endogeneity vs Identification

Endogeneity causes lack of identifiability. In a regression models we need some form of orthogonality between errors and regressors to rule out endogeneity

Common causes of endogeneity

1. **Omission of relevant variables** (seen in Part I)
   Possibile solutions: include the missed variable, find a good proxy variable, find an IV (see later),
2. **Errors-in-variables (aka measurement errors)**
   regressors are measured with errors. Possibile solutions: get a more accurate measurement, use a proxy or IV (see later)
3. **Simultaneous causality**
   When $Y = \beta_1 + \beta_2 X$ and $X = \theta_1 + \theta_2 Y$ are both true, i.e. X causes Y and vice versa (e.g. Y=quantity of oil, X=price of oil).
   Possibile solutions: use IV, *randomized controlled experiments*

Notes

Exogeneity

$A \to B$ stands for: "A causes B in a linear way". The simple linear model

$$Y = \beta Z + \varepsilon$$

is represented as

$$\begin{array}{ccc} Z & \to & Y \\ & \nearrow & \\ \varepsilon & & \end{array}$$

The regressor Z is exogenous = orthogonal to the error. The error should incorporate exogenous factors that produce variations of Y that can be distinguished from the variations caused by Z.

The ultimate goal is to use β to measure the impact of Z on Y all else equal. This is only achieved if there is no "linear" link between Z and ε.

Notes

Endogeneity mechanism

Suppose Z is endogenous

$$\begin{array}{ccc} Z & \rightarrow & Y \\ \uparrow & \nearrow & \\ \varepsilon & & \end{array}$$

Now ε correlates with Z. Therefore, when ε moves, also Z moves for the same reason!

This implies that β is not able to isolate the impact of Z on Y. In other words the role of β is not identified.

Notes

Instrumental Variables (IV) "filtering" mechanism

Now suppose that there exists a third variable X

$$\begin{array}{ccccc} X & \rightarrow & Z & \rightarrow & Y \\ & & \uparrow & \nearrow & \\ & & \varepsilon & & \end{array}$$

X is called Instrumental Variable (IV) if it only causes Z, with an indirect effect on Y.

The causation mechanism is as follow

- when ε moves, this also produces variation in Z
- but these variations are recognized, because we known that Z moves on its own if X moves
- if X is a fairly good linear predictor of Z (e.g. $\mathrm{Cov}[X, Z] \neq 0$), observing X we know whether Z is varying for reasons not connected to ε.

Therefore, we know how Z impacts $Y \implies \beta$ is identified

Notes

Identification of the scalar regression model using IVs

Assume

- the regression model of interest is $Y = \beta Z + \varepsilon$
- Z is endogenous, that is $\mathrm{Cov}[Z, \varepsilon] \neq 0$
- we can observe a variable X such that $\mathrm{Cov}[X, \varepsilon] = 0$ and $\mathrm{Cov}[X, Z] \neq 0$
- we know how to compute moments from the joint distribution of the observables (Y, Z, X)

β is identified

from the joint distribution of the observables (Y, Z, X), we can compute

$$\mathrm{Cov}[X, Y] = \mathrm{Cov}[X,\ \beta Z + \varepsilon] = \beta\, \mathrm{Cov}[X, Z] + \mathrm{Cov}[X, \varepsilon]$$

By assumption $\mathrm{Cov}[X, Z] \neq 0$, and $\mathrm{Cov}[X, \varepsilon] = 0$, therefore there is a one-to-one map between β and the population distribution, that is

$$\beta = \frac{\mathrm{Cov}[X, Y]}{\mathrm{Cov}[X, Z]}$$

Notes

General conditions for IVs

X is a *Valid Instrument* of an endogenous regressor Z if it fulfills:

1. exogeneity: IVs are orthogonal to the regression error
2. relevance: instruments are non orthogonal to the regressors.

Scalar regression example

$Y = \beta Z + \varepsilon$, with $\mathrm{Cov}(X, \varepsilon) = 0$ and $\mathrm{Cov}(X, Z) \neq 0$:

- both validity conditions are fulfilled
- the relevance rules out the possibility that X causes Y directly, that would be if X was a regressor of Y. In fact, if

  $$Y = \beta Z + \delta X + u$$

  then $\varepsilon = \delta X + u \implies \mathrm{Cov}(\varepsilon, X) \neq 0$ violating the validity assumption. In this case validity is fulfilled if $\delta = 0$
- Relevance assumption implies that $Z = \gamma_1 + \gamma_2 X + \eta$ with $\gamma_2 \neq 0$

Notes

An exogenous regressor can be *instrument* by itself

- we have a vector of regressors $\boldsymbol{z}$, but some of them are endogenous
- partition $\boldsymbol{z} = (\boldsymbol{z}_1, \boldsymbol{z}_2)$, suppose we put endogenous regressors in $\boldsymbol{z}_1$
- suppose the vector $\boldsymbol{x}_1$ contains *valid* IVs for $\boldsymbol{z}_1$

Then $\boldsymbol{x} = (\boldsymbol{x}_1, \boldsymbol{z}_2)$ is a vector of *valid* IV for $\boldsymbol{z}$

Finding IVs is Art!

Exogeneity

can't be checked because involves the unobservable error. What to do? Use scientific creativity and economic intuition to convincingly argue that IV are not endogenous.

Relevance

for vectors of regressors this is more difficult to formalize and interpret. There exist methods to test it. However, the methodology is not general and depends on the context.

Notes

Example: returns to schooling

Consider the following linear model:

$$\log(\texttt{wage}_i) = \beta_1 + \beta_2 \texttt{education}_i + \beta_3 \texttt{ability}_i + \varepsilon_i \qquad \text{(Model 1)}$$

with both regressors being exogenous.

`ability` is not observable, but we are interested in estimating the marginal impact of `education` on `wage`. We consider

$$\log(\texttt{wage}_i) = \beta_1 + \beta_2 \texttt{education}_i + u_i \qquad \text{(Model 2)}$$

- we are omitting a relevant variable. Now $\texttt{ability}_i$ goes into the error. In fact: $u_i = \beta_3 \texttt{ability}_i + \varepsilon_i$
- but $\mathrm{Cor}(\texttt{education}_i, \texttt{ability}_i) \neq 0 \implies \mathrm{Cor}(\texttt{education}_i, u_i) \neq 0$ which means that in (Model 2) the `education` becomes endogenous.
- An estimator of β_2 won't be consistent, because β_2 is not identified.

Notes

We want to use our data to measure the impact of education on wages. If we find a good IV for `education`, we can still use (Model 2) to identify β_2.

- $X =$"*number of days from 1/1/0 to birth date*".
  Certainly $\mathrm{Cor}[X, \texttt{ability}] = 0$ so $\mathrm{Cor}[X, u] = 0$, however $\mathrm{Cor}[X, \texttt{education}] = 0$... X doesn't work!
- $X =$"*mother's education*".
  Certainly $\mathrm{Cor}[X, \texttt{education}] > 0$. However, $\mathrm{Cor}[X, \texttt{ability}] > 0$ implying $\mathrm{Cor}[X, u] > 0$ too... X doesn't work!
- $X =$"*number of siblings (total of sisters + brothers)*".
  It turns out that $\mathrm{Cor}[X, \texttt{education}] < 0$. And $\mathrm{Cor}[X, \texttt{ability}] = 0$ implying $\mathrm{Cor}[X, u] = 0$... well done!

We can use $X =$"*number of siblings*" as an instrument for `education` in (Model 2), and get β_2 identified. Now it make sense to estimate β_2 from (Model 2)

Notes

Proxy vs IV

A proxy variable is a regressor that is almost perfectly correlated with a variable that can't be sampled. For instance $\texttt{IQ} = \gamma\texttt{ability}$.

$$\begin{aligned}\log(\texttt{wage}_i) =& \beta_1 + \beta_2\texttt{education}_i + \beta_3\texttt{ability}_i + \varepsilon_i \\ =& \beta_1 + \beta_2\texttt{education}_i + \frac{\beta_3}{\gamma}\texttt{IQ}_i + \varepsilon_i \qquad \text{(Model 3)}\end{aligned}$$

$\boldsymbol{b}$ = least square estimator for (Model 3). Now b_3 is consistent for (β_3/γ), and b_2 is consistent for β_2. We are not able to identify β_3 and γ separately. If we are interested in effects of educational policy we want to get β_2 right.

- X is a proxy for Z: we require that X is a function of Z. Then, eventually after a rescaling, X may well be a regressor of Y.
- X is an IV for Z: we require that X does not cause Y directly, therefore X can't be a regressor of Y (see previous point)

Notes

IV strength

The validity condition is satisfied when IV are non orthogonal to the regressors. But X and Z are non orthogonal even if $\mathrm{Cor}[X, Z]$ is modest

In order to work *efficiently*, an IV should be a fairly strong predictor for the endogenous regressors. Recall: the role of the IV is to let us see when the endogenous regressor is moving on its own.

Weak instruments

IVs are strong predictors for the endogenous regressors when they explain much of their variations. If this does not happen, we say that X is a *weak instrument*.

Weak IVs can cause a lot of trouble!

Notes

Notes